

CTRL: A Joint Cluster Head Selection, Task Offloading, and UAV Redeployment Scheme for Load-Balanced FANETs

Author Name

Abstract—

Index Terms—

I. INTRODUCTION

The rapid growth of the uses of unmanned aerial vehicles (UAV) in broader domains including agriculture, military operations, land surveying, disaster management, and other emerging fields for UAV applications. However for the seamless and efficient uses of UAVs manual control operations are often insufficient for utilize full functionality of UAVs, so UAVs are required to have fully automatic control so that full functionality is used efficiently and provides satisfactory performance to end users. Users are frequently generating data which are retrieving and sending in a time period which causes network overload. So getting rid of this challenge Mobile edge computing (MEC) is deployed at Unmanned aerial vehicle(UAV) which alleviate the task by providing computation offloading on edge node and provides low latency to end user for a better reliability and connectivity [1]. In base stations and UAVs both have computing power for the computation, can assigned to offload the UAV task. For UAV task offloading it is difficult to determine how the number clustering center for balancing between energy consumption of UAVs and the clustering centers [2]. Controlling UAV swarm network is tedious task without of central authority, hence author in [3] proposed a cluster head in hierarchical UAV swarm network act as a communication and coordination centers for other UAVs in a swarm. In wireless sensor networks(WSNs) the sensor nodes(SNs) continuously communicate with nearby SNs, which causes rapid energy depletion. So a cluster head is required to have a better co-ordination between Sensor nodes for load balancing and optimization among sensor nodes to minimize the depletion of energy flows [4].

Unmanned Aerial Vehicle(UAVs) are widely used in various real-time applications such as disaster management,surveillance,military operation,package delivery because of UAVs are capable of easy deployment,flexibility and mobility.In past few years,UAV swarm network have gained popularity to collaboratively performs complex task efficiently than a uni UAV. With compact sizes of UAVs have limited computation power,storage capacity and battery. UAVs form swarm of network but managing and co-ordinating among them is tedious task. To improve the co-ordination among UAVs,clustering techniques are mainly used for dividing the UAVs into multiple clusters.In each of a clusters there contains

a Cluster Head(CH) which have primarily task to manage task coordination,communication management to nearby cluster head of other clusters,and resource allocation among different UAVs.The cluster based framework reduces overhead in communication,network scalability improves and efficient utilization of UAVs are done.Along with it cluster head also enables the inter-cluster communication and provides assisting in stability of workload distribution efficiently. In static UAV environments, task demand not vary continuously but in dynamic environment,task demand vary continuously across the different clusters.In dynamic some of the cluster have utilized fully(overloaded) due to heavily task requests while other donot utilized fully.This imbalance causes increase in processing delay,communication overhead,energy degradation which degraded the overall network performance drastically. The solution of this kind of problem is UAV redeployment among clusters.In this type of strategy, the least used UAV from that cluster can be transferred to overloaded cluster to facilitate task execution and balance the load . However,choosing suitable UAVs for transefing from least overloaded to overloaded cluster is a tedious task due to dynamic network conditions,limited energy resources, and mobility of UAVs and users on ground.For smooth coordination among different clusters to figure out which UAV migrates, in what situation migration performs and how the task distribution divides efficiently. For UAV redeployment and load balancing a game theoretic non cooperative approach is followed.

A. Motivation

Unmanned Aerial Vehicles(UAVs) are widely used in many real world applications because of small sizes of UAVs ,such as disaster management,surveillance and package delivery from warehouses to home.In spite of small size of UAVs which forms swarm UAV network, is a complex task because of network topology changes,limited resource and frequent movement of UAVs.To tackle these challenges multiple cluster forms,where each cluster have dedicated Cluster Head(CH) which manages UAVs in that particular cluster for resource sharing,user reassignment and energy efficiently. With such cluster system formation by UAV ,Cluster Head also communicate with other cluster to provide facility to redeployment of UAVs from one to another at the time of addition UAV needed when that particular cluster cannot handle with existing UAV causes can be limited resources or load has increased drastically in dynamic environment. Therefore, combining a

clustering mechanism with a non-cooperative game-theoretic approach gives an effective and reliable solution for redeployment of UAV in swarm of network. By integrating those two things scalability, reliability improves in UAV swarm networks.

B. Research Challenges

C. Contribution

II. RELATED WORK

A. UAV Applications

B. UAV-as-a-Service

C. Research Gap

III. PROBLEM DESCRIPTION

A. Problem Scenario

B. System Architecture

Fig. 1: Proposed CTRL system architecture.

C. Network Topology

Creating a Cartesian coordinate system of a three-dimension system. $\mathcal{N} = \{1, 2, \dots, N\}$ of UAVs hovers at a unchangeable altitude H (m) above the ground, providing its service to set $\mathcal{G} = \{1, 2, \dots, G\}$ of User Equipments (UEs) operating by users on ground. In individual time interval $t \in \mathcal{T} = \{1, \dots, T\}$, the 3-D positioning of the UAV n is

$$d_{im} = \|\mathbf{a}_i - \mathbf{a}_m\| = \sqrt{(x_i - x_m)^2 + (y_i - y_m)^2 + (h_i - h_m)^2}. \quad (1)$$

$$\mathbf{q}_n(t) = (x_n(t), y_n(t), H), \quad n \in \mathcal{N}. \quad (2)$$

and UE position g is $\mathbf{s}_g(t) = (x_g(t), y_g(t), 0)$. The UAVs makes a swarm so partitioned into K disjoint clusters $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_K\}$ such that $\bigcup_{k=1}^K \mathcal{C}_k = \mathcal{N}$ and $\mathcal{C}_k \cap \mathcal{C}_{k'} = \emptyset$ for $k \neq k'$. Except cluster $\mathcal{C}_k$, exactly one among from all the UAVs works as the *cluster head* (CH) and the all others are *cluster members* (CMs).

D. Air-to-Air Communication Model

1) *UAV-to-UAV (Intra-Cluster) A2A Channel*: in spite of all UAVs control at identical altitude H , the communication link inside intra-cluster are primarily line-of-sight (LoS). cluster member m from UAV i transmitting with power p_i with received signal-to-interference plus-noise ratio (SINR) is

$$\text{SINR}_{im} = \frac{p_i g_{im} d_{im}^{-\eta}}{\sigma^2 + \sum_{j \neq i} p_j g_{jm} d_{jm}^{-\eta}} \quad (3)$$

where d_{im} express the inter-UAV distance, g_{im} express the channel gain, η is the path-loss exponent, σ^2 is the noise power, and p_i is UAV transmit power i .

For Line of Sight (LoS) UAV-to-UAV communication, the path-loss exponent usually ranges from 2 to 2.5 is taken from [8]. In this work, η is borrowed.

The gain associated with small-scale fading gain adhere to Rayleigh fading framework and is represented as

$$g_{im} \sim \text{Exp}(1). \quad (4)$$

The thermal noise power is

$$\sigma^2 = F k_B T_{\text{env}} B. \quad (5)$$

where F is noise from receiver side, k_B is standard Boltzmann's constant, T_{env} is the environment temperature, and B is bandwidth of the entire channel is adopted from [8].

The transmit power satisfies

$$0 \leq p_i \leq P_i^{\max}. \quad (6)$$

where $P_i^{\max}$ signifies the maximum permissible transmission power of the UAV i .

a) *Small-scale fading gain g_{im}* : The received challenging baseband is

$$\tilde{y}_{im} = \sum_{\ell=1}^L a_{\ell} e^{j\phi_{\ell}} s(t - \tau_{\ell}). \quad (7)$$

where a_{ℓ} , $\phi_{\ell} \sim \mathcal{U}[0, 2\pi)$, and τ_{ℓ} are the ℓ -th multipath amplitude, phase, and delay. For L sufficiently large, the Central Limit Theorem gives $\text{Re}[\tilde{y}_{im}]$, $\text{Im}[\tilde{y}_{im}] \sim \mathcal{N}(0, \sigma_h^2/2)$, so the power gain

$$g_{im} = |\tilde{y}_{im}|^2 \stackrel{d}{=} \text{Exp}(1), \quad \mathbb{E}[g_{im}] = 1 \quad (8)$$

Rayleigh power follow a average channel condition which is 1 in Exp. it is a standard worst case for LoS UAV when Ricin K-factor is not known [8].

b) *Transmit power p_i* : The power $p_i \in [0, P_i^{\max}]$ is the strategy varying of UAV i for electing CH on applying election game. Its maximum power limit for transmitting P_i^{hw} and the rest energy during a time interval:

$$P_i^{\max} = \min\left(P_i^{\text{hw}}, \frac{E_i^{\text{res}}}{\Delta t}\right) \quad (9)$$

where E_i^{res} (J) is examined by the preinstalled coulomb counter and Δt (s) is the time-interval.

2) *Time-slot duration Δt* : The channel assumption required g_{im} is given by quasi-static to stay steady within each particular slots. The channel coherence time is given by the Jakes model is

$$T_c = \frac{9c}{16\pi v_i^{\max} f_c} \quad (10)$$

where c is speed of light, $v_i^{\max}$ is the maximum UAV speed, and f_c is the carrier frequency. The outcome follows from the level-crossing-rate analysis of the Rayleigh, which give in Doppler Bandwidth $f_D^{\max} = v_i^{\max} f_c / c$, and hence $T_c \approx 9 / (16\pi f_D^{\max})$ [9] The time duration must satisfy

$$\Delta t \leq T_c \quad \text{and} \quad \Delta t \geq T_{\text{meas}} + T_{\text{proc}} \quad (11)$$

where T_{meas} is the pilot-symbol overhead time spent during sending it and T_{proc} is the KKT optimization after getting channel information

E. SINR-Based Coverage Probability

The power of all the UAVs are $\mathbf{p} = (p_1, \dots, p_{N_k})$, the probability that UAV i ' broadcast can decoded by CM (i.e., $\text{SINR}_{im} \geq \gamma_{\min}$) is obtained by averaging (3) over the Rayleigh fading distribution (8). Since $g_{im} \sim \text{Exp}(1)$, we have

$$\begin{aligned} \Pr[\text{SINR}_{im} \geq \gamma_{\min}] &= \Pr\left[g_{im} \geq \frac{\gamma_{\min}(\sigma^2 + I_{im})}{p_i d_{im}^{-\eta}}\right] \\ &= \mathbb{E}_{I_{im}} \left[\exp\left(-\frac{\gamma_{\min} \sigma^2 d_{im}^{\eta}}{p_i}\right) \cdot \prod_{j \neq i} \frac{p_i d_{im}^{-\eta}}{p_i d_{im}^{-\eta} + \gamma_{\min} p_j d_{jm}^{-\eta}} \right] \end{aligned} \quad (12)$$

where the expectation uses the Laplace transform of $g_{jm} \sim \text{Exp}(1)$ for each interfering UAV j . The expected coverage fraction of all CMs by UAV i is

$$\mathbb{E}[C_i(p_i, \mathbf{p}_{-i})] = \frac{1}{N_k - 1} \sum_{\substack{m \in \mathcal{C}_k \\ m \neq i}} \Pr[\text{SINR}_{im} \geq \gamma_{\min}] \quad (13)$$

1) *Derivation of $\gamma_{\min}$* : The $\gamma_{\min}$ is not assumed as a constant; it is obtained from the minimum rate required for successful beacon/control-message decoding during the initial CH election stage. When UAVs are scattered, each UAV broadcasts a beacon message to estimate how many neighbouring UAVs can be reliably reached. The required spectral efficiency is determined by the beacon descriptor as

$$R_{\text{req}} = \frac{L_b}{T_b \cdot B_c} \quad (\text{bits/s/Hz}). \quad (14)$$

where L_b is the beacon/control packet size in bits, T_b is the maximum allowable beacon decoding time, and B_c is the control-channel bandwidth.

Using the Shannon rate relation, the achievable spectral efficiency of the UAV-to-UAV control link is expressed as

$$R_{im} = \log_2(1 + \text{SINR}_{im}). \quad (15)$$

For reliable decoding, the achievable spectral efficiency must satisfy $R_{im} \geq R_{\text{req}}$. Therefore, the minimum SINR threshold required for successful beacon decoding is obtained as

$$\gamma_{\min} = 2^{R_{\text{req}}} - 1 = 2^{\frac{L_b}{T_b \cdot B_c}} - 1. \quad (16)$$

Thus, $\gamma_{\min}$ is a system-derived parameter that reflects the minimum communication quality required for reliable control-message exchange. A UAV-to-UAV link from candidate CH i to member UAV m is considered reachable only when

$$\text{SINR}_{im} \geq \gamma_{\min}. \quad (17)$$

F. Health Index H_i

The overall health of a UAV is shown by $H_i \in [0, 1]$ which examines three independent and non-overlapping aspects of the UAVs operating condition: energy, thermal state, and link quality for communication

1) *Energy health h_i^E* : Since battery tolerance is a major factor for UAV operation, the health component is derived from the battery behaviour. Most of the research UAVs used Lithium-Polymer (LiPo) batteries, whose voltage performs a nonlinear relationship with charge state. Shepherd electrochemical battery model accurately obtained [10]:

$$V_{\text{batt}}(q) = E_0 - K \frac{Q}{Q - q} + A e^{-Bq} \quad (18)$$

where E_0 (V) is the open-circuit voltage, K (V/Ah) the polarisation constant, Q (Ah) the rated capacity, A (V) the exponential amplitude, B (1/Ah) the exponential time constant, and q (Ah) the extracted charge—all from the IEC 61960 discharge test. The true *usable* energy remaining from charge state q down to the hardware cutoff voltage V_{cut} is

$$\begin{aligned} E_i^{\text{usable}}(q) &= \int_q^Q [V_{\text{batt}}(q') - V_{\text{cut}}] dq' \\ &= (E_0 - V_{\text{cut}})(Q - q) - KQ \ln \frac{Q}{Q - q} + \frac{A}{B} (1 - e^{-B(Q - q)}) \end{aligned} \quad (19)$$

The normalised usable energy gives energy health:

$$h_i^{E, \text{exact}}(q) = \frac{E_i^{\text{usable}}(q)}{E_i^{\text{usable}}(0)}. \quad (20)$$

For ease of analysis in the KKT derivation, approximate $h_i^{E, \text{exact}}$ by a smooth exponential $1 - e^{-\alpha_E s}$ with $s = E_i^{\text{res}}/E_i^{\text{max}} \in [0, 1]$. The parameter α_E is determined by minimising the L^2 approximation error:

$$\min_{\alpha_E \geq 0} \int_0^1 [h_i^{E, \text{exact}}(s) - (1 - e^{-\alpha_E s})]^2 ds \quad (21)$$

Setting the derivative with respect to α_E to zero yields the implicit optimality condition

$$\int_0^1 h_i^{E, \text{exact}}(s) \cdot s e^{-\alpha_E s} ds = \int_0^1 (1 - e^{-\alpha_E s}) \cdot s e^{-\alpha_E s} ds, \quad (22)$$

$$h_i^E = 1 - \exp\left(-\alpha_E \frac{E_i^{\text{res}}}{E_i^{\text{max}}}\right). \quad (23)$$

2) *Thermal health h_i^T* : The onboard thermal sensor provides the current CPU temperature T_i^{curr} (K). since increase in temperature can influence performance throttling, thermal health is obtained by normalizing T_i^{curr} and T_i^{min} and the maximum acceptable temperature T_i^{max} given by

$$h_i^T = 1 - \frac{T_i^{\text{curr}} - T_i^{\text{min}}}{T_i^{\text{max}} - T_i^{\text{min}}} \in [0, 1] \quad (24)$$

The CPU frequency of UAV i after inspecting thermal throttling is obtained by:

$$f_i^{\text{res}} = f_i^{\text{max}} \cdot h_i^T \quad (25)$$

This establishes a one-way dependency from the current temperature to the thermal health and then to the residual CPU frequency, expressed as

$$T_i^{\text{curr}} \rightarrow h_i^T \rightarrow f_i^{\text{res}}$$

Therefore, the thermal-health component is independent of the energy-health component (h_i^E), since (h_i^E) is calculated only from the residual battery energy E_i^{res} .

3) *Link-quality health* h_i^L : The Received Signal Strength Indicator (RSSI) is measure of how well a signal can be received over time, which is different from the immediate signal strength in each slot, called per slot channel chain g_{im} . time-averaged reachability metric, Let RSSI_{im} be the smoothed RSSI value measure from UAV(m) to UAV(i), and let $(\text{RSSI}_{\min} = -90 \text{ dBm})$ be the lowest RSSI level that the hardware can still decode.

$$h_i^L = \frac{1}{|\mathcal{C}_k| - 1} \sum_{\substack{m \in \mathcal{C}_k \\ m \neq i}} \mathbf{1}[\text{RSSI}_{im} \geq \text{RSSI}_{\min}] \in [0, 1] \quad (26)$$

4) *Composite index*: The three dimensions are combined via a weighted geometric mean:

$$H_i = (h_i^E)^{w_E} (h_i^T)^{w_T} (h_i^L)^{w_L} \quad (27)$$

with $w_E + w_T + w_L = 1$. The geometric (rather than arithmetic) mean ensures that $H_i \rightarrow 0$ whenever *any* dimension collapses, reflecting the fact that a UAV with dead batteries, a throttled CPU, or an isolated radio link is ineligible to serve as CH regardless of its other metrics. Weights $(w_E, w_T, w_L) = (0.5, 0.25, 0.25)$ are derived via Saaty's Analytic Hierarchy Process with consistency ratio $\text{CR} < 0.1$.

Non-overlap summary: h_i^E uses E_i^{res} ; h_i^T uses T_i^{curr} (different physical sensor); h_i^L uses RSSI (time-averaged link metric, independent of the per-slot g_{im} in (8)).

IV. CLUSTER HEAD ELECTION GAME

A. Problem Formulation

Within cluster $\mathcal{C}_k$ (cardinality $N_k = |\mathcal{C}_k|$), every UAV competes for the CH role. The CH needs to send coordination messages to every other UAVs.CMs; hence the natural strategy variable in the coordination broadcast power $p_i \in [0, P_i^{\max}]$. every UAV competes for the CH role. The CH needs to send coordination messages to every other UAVs.CMs; hence the natural strategy variable in the coordination broadcast power $p_i \in [0, P_i^{\max}]$. A high value of p_i leads to broader coverage but uses more energy. A UAV that is running low on power or has slowed down because of heat will naturally choose, low power, analytically self-excluding without any externally imposed threshold.

Definition 1 (CH Election Game). *The CH election game is the triple $\mathcal{G}_{\text{CH}} = \langle \mathcal{C}_k, \{[0, P_i^{\max}]\}_{i \in \mathcal{C}_k}, \{\pi_i\}_{i \in \mathcal{C}_k} \rangle$, where $\pi_i : \prod_{j \in \mathcal{C}_k} [0, P_j^{\max}] \rightarrow \mathbb{R}$ is the payoff of player i .*

B. Payoff Function

The payoff of UAV i balances coverage gain against energy expenditure, weighted by its health index:

$$\pi_i(p_i, \mathbf{p}_{-i}) = H_i \mathbb{E}[C_i(p_i, \mathbf{p}_{-i})] - \frac{(1 - H_i) p_i \Delta t}{E_i^{\text{res}}} \quad (28)$$

a) *Interpretation*: When $H_i \rightarrow 1$: the UAV is healthy, so coverage gain dominates and the energy cost term vanishes; the UAV competes aggressively. When $H_i \rightarrow 0$: the energy-cost coefficient $\rightarrow (E_i^{\text{res}})^{-1} \cdot \Delta t$, making π_i strictly decreasing in p_i , so the optimal power is $p_i^* = 0$ —the UAV self-excludes without any threshold check. The weight $(1 - H_i)/E_i^{\text{res}}$ is the marginal energy cost per unit power, derived from the chain rule: $\partial \pi_i / \partial E_i^{\text{cost}} = -(1 - H_i)/E_i^{\text{res}}$, where $E_i^{\text{cost}} = p_i \Delta t$.

C. KKT Optimality Conditions

Each UAV i solves the individual optimisation:

$$\max_{p_i} \pi_i(p_i, \mathbf{p}_{-i}) \quad \text{s.t.} \quad 0 \leq p_i \leq P_i^{\max}, \quad p_i \Delta t \leq E_i^{\text{res}} \quad (29)$$

The Lagrangian is

$$\mathcal{L}_i = \pi_i + \mu_i^{(1)} p_i - \mu_i^{(2)} (p_i - P_i^{\max}) - \mu_i^{(3)} (p_i \Delta t - E_i^{\text{res}}) \quad (30)$$

with $\mu_i^{(1)}, \mu_i^{(2)}, \mu_i^{(3)} \geq 0$.

1) *Stationarity*: Setting $\partial \mathcal{L}_i / \partial p_i = 0$:

$$H_i \cdot \left. \frac{\partial \mathbb{E}[C_i]}{\partial p_i} \right|_{p_i^*} = \frac{1 - H_i}{E_i^{\text{res}}} + \mu_i^{(2)} + \mu_i^{(3)} \Delta t - \mu_i^{(1)} \quad (31)$$

Computing $\partial \mathbb{E}[C_i] / \partial p_i$ analytically, define $A_{im} = \gamma_{\min} \sigma^2 d_{im}^\eta$ and $B_{jm} = \gamma_{\min} p_j d_{jm}^{-\eta} d_{im}^\eta$. Then the gradient of the m -th coverage term is

$$\begin{aligned} \frac{\partial}{\partial p_i} \Pr[\text{SINR}_{im} \geq \gamma_{\min}] &= \frac{A_{im}}{p_i^2} e^{-A_{im}/p_i} \prod_{j \neq i} \frac{p_i d_{im}^{-\eta}}{p_i d_{im}^{-\eta} + \gamma_{\min} p_j d_{jm}^{-\eta}} \\ &+ e^{-A_{im}/p_i} \sum_{j \neq i} \frac{B_{jm}/p_i^2}{(1 + B_{jm}/p_i)^2} \prod_{\ell \neq i, \ell \neq j} (\dots) \end{aligned} \quad (32)$$

where the remaining product terms are bounded and positive.

2) *Complementary slackness*:

$$\mu_i^{(1)} p_i^* = 0, \quad \mu_i^{(2)} (p_i^* - P_i^{\max}) = 0, \quad \mu_i^{(3)} (p_i^* \Delta t - E_i^{\text{res}}) = 0 \quad (33)$$

Three cases arise:

Case A (interior, energy abundant): All factors are set to zero; (31) simplifies to condition for the equilibrium expressed analytically as

$$H_i \cdot \left. \frac{\partial \mathbb{E}[C_i]}{\partial p_i} \right|_{p_i^*} = \frac{1 - H_i}{E_i^{\text{res}}} \quad (34)$$

which means that the increase in outermost coverage is equal to the normalized cost of outermost energy. The Equation (34) is solved for p_i^* using the method of bisection, as assured by the strict-concavity argument above.

Case B (energy-constrained): Since $\mu_i^{(3)} > 0$, so the optimal power level for the UAV is $p_i^* = E_i^{\text{res}} / \Delta t$ (battery fully consumed in one slot). A UAV with low-energy is automatically given a lower power setting as given by the KKT condition.

Case C (withdrawal): The value of $\mu_i^{(1)} > 0$, so $p_i^* = 0$. The UAV does not broadcast; its Utility Score score is zero and it is excluded from the CH election without any threshold set by outside.

D. Equilibrium Utility Score and CH Selection

The *Utility Score* of UAV i at the Nash equilibrium $\mathbf{p}^* = (p_1^*, \dots, p_{N_k}^*)$ is

$$b_i = \pi_i(p_i^*, \mathbf{p}_{-i}^*) = H_i \cdot \mathbb{E}[C_i(p_i^*, \mathbf{p}_{-i}^*)] - \frac{(1 - H_i) p_i^* \Delta t}{E_i^{\text{res}}}. \quad (35)$$

The cluster head is elected as

$$i^* = \arg \max_{i \in \mathcal{C}_k, p_i^* > 0} b_i \quad (36)$$

UAVs with $p_i^* = 0$ (Case C) are automatically excluded.

E. Theoretical Guarantees

Existence of Pure-Strategy Nash Equilibrium: The game for electing a cluster head, denoted by $\mathcal{G}_{\text{CH}}$, has a minimum of one pure strategy Nash equilibrium (NE). As per the Debreu Glicksberg fan theorem, a pure strategy NE is assured if the following criteria met. First the set of strategies $[0, P_i^{\text{max}}]$, is non empty, compact and convex for every UAV i . Second the payoff function $\pi_i(p_i, p_{-i})$ is continuous for both p_i and p_{-i} and p_{-i} shows quasi concavity in p_i for fixed p_{-i} . In the game defined that the payoff function is made up of exponential, product, and related components, which guarantees its continuity. Additionally, Lemma V-E confirms the quasi-concavity of π_i with reference to p_i . Thus all the necessary criteria satisfied, so that the game $\mathcal{G}_{\text{CH}}$ highlights at least one pure strategy NE.

Strict Concavity: For a fixed strategy configuration of other UAVs, denoted as p_{-i} , is the utility function $\pi_i(p_i, p_{-i})$ performs strict concavity concerning p_{-i} within the fixed range $(0, P_i^{\text{max}}]$. The element associated with energy expenditure is

$$-\frac{(1 - H_i) p_i \Delta t}{P_i^{\text{max}}}$$

is linear in p_i , and therefore its second derivative equals to zero. The exponential factor connected to co

$$\frac{\partial^2}{\partial p_i^2} \exp\left(-\frac{A_{im}}{p_i}\right) = \left(\frac{A_{im}}{p_i^2}\right) \left(\frac{A_{im}}{p_i^2} - \frac{2}{p_i}\right) \exp\left(-\frac{A_{im}}{p_i}\right).$$

The value becomes negative when $p_i > A_{im}/2$, which relates to the realistic working range at which the probability of coverage is high. Furthermore, the log concavity is connected with the exponential elements helps to preserve the concaveness of the multiplicative character of coverage probability. Thus

$$\frac{\partial^2 \pi_i}{\partial p_i^2} < 0$$

which proves that $\pi_i(p_i, p_{-i})$ is strictly concave in p_i .

Analytical Self-Exclusion of Energy-Depleted UAVs: If the health indicator of UAV i near zero, i.e., $H_i \rightarrow 0$, then ideal transmission power also goes towards zero, $p_i^* \rightarrow 0$, and as a result its election Utility Score becomes zero, $b_i \rightarrow 0$. This implies that a UAV with fully depleted energy is automatically eliminated from the cluster-head election process.

When $H_i \rightarrow 0$, the payoff function reduces to

$$\pi_i(p_i, p_{-i}) \rightarrow -\frac{p_i \Delta t}{P_i^{\text{max}}}$$

which is strictly decreasing in p_i . Therefore, the best answer is found at lower limit of the possible set, i.e.,

$$p_i^* = 0$$

In the KKT formulation, turns on the lower bound multiplier $\mu_i^{(1)} > 0$. From the Utility Score formula, then related Utility Score meets

$$b_i = 0$$

Thus, an energy drained UAV cannot become a cluster head.

Uniqueness of NE under Weak Interference Coupling: The Nash equilibrium of $\mathcal{G}_{\text{CH}}$ is unique when the interference between players is not too strong. This condition is shown by:

$$\kappa \triangleq \max_i \max_{j \neq i} \left| \frac{\frac{\partial^2 \pi_i}{\partial p_i \partial p_j}}{\frac{\partial^2 \pi_i}{\partial p_i^2}} \right| < 1. \quad (37)$$

Under the condition $\kappa < 1$, the best-response mapping

$$BR_i(p_{-i}) = \arg \max_{p_i \in [0, P_i^{\text{max}}]} \pi_i(p_i, p_{-i})$$

It becomes a contraction mapping when using the ℓ_∞ norm. So because of the Branch fixed point theorem, the iterative best response update by

$$p_i^{(t+1)} = BR_i(p_{-i}^{(t)})$$

converges to a unique fixed point p^* . Because any point that stays the same after applying the best response mapping is a Nash equilibrium. In this game, p^* is the only Nash equilibrium.

In the proposed UAV clustering scenario, the weak connection condition is satisfied when the cluster density is bounded as

$$\frac{N_k}{\rho_k^2} < \frac{2\pi\eta^2\sigma^2}{P_i^{\text{max}}g_0}$$

In cluster k , N_k represents the number of UAVs, ρ_k is cluster size, η shows how signal strength decreases with distance, σ^2 is the background noise level, and g_0 is the base value for the communication channel. This condition means the cluster shouldn't be too loaded. In real world scenario UAV swarms, this condition met when cluster size is chosen close to the communication range, i.e., $\rho_k = R_{\text{comm}}$

F. Initial Power and Convergence

Before the first iteration, UAV i sets its initial broadcast power without knowledge of $\mathbf{p}_{-i}$:

$$p_i^{(0)} = H_i \cdot \frac{E_i^{\text{res}}}{\Delta t} \quad (38)$$

Justification: From energy conservation, $E_i^{\text{res}}/\Delta t$ is the maximum sustainable power for one slot. Scaling by H_i ensures that a healthy UAV broadcasts near its energy limit (high candidacy signal) while a depleted UAV broadcasts near zero (low candidacy). No other UAV's state is required.

Each UAV broadcasts $p_i^{(t)}$ in a pilot beacon (2 bytes overhead), receives $\{p_j^{(t)}\}_{j \neq i}$, and updates via $p_i^{(t+1)} = BR_i(\mathbf{p}_{-i}^{(t)})$. Convergence to ε -accuracy requires

$$T_{\text{conv}} = \left\lceil \frac{\log \varepsilon}{\log \kappa} \right\rceil. \quad (39)$$

Algorithm 1 CH Election Game

Require: $C_k = \{u_1, u_2, \dots, u_{N_k}\}$, E_i^{res} , T_i^{curr} , RSSI_{im} , p_i^{max} , $\gamma_{\min}$, f_i^{res}

Ensure: Elected cluster head i^*

- 1: Derive $\gamma_{\min}$ from beacon spectral-efficiency requirement
- 2: Compute slot duration Δt from channel coherence time
- 3: Set p_i^{max} from hardware limit and residual energy, $\forall i \in C_k$
- 4: **for** each UAV $u_i \in C_k$ **do**
- 5: Compute energy health h_i^E from battery discharge model
- 6: Compute thermal health h_i^T from CPU temperature reading
- 7: Compute link-quality health h_i^L from neighbour RSSI reachability
- 8: Compute composite health index H_i via weighted geometric mean
- 9: Initialise broadcast power $p_i^{(0)}$ scaled by H_i and E_i^{res}
- 10: **end for**
- 11: $t \leftarrow 0$
- 12: converged $\leftarrow$ false
- 13: **while** converged = false **do**
- 14: Each u_i broadcasts $p_i^{(t)}$ in pilot beacon
- 15: Each u_i receives $p_j^{(t)}$, $j \neq i$, from cluster members
- 16: **for** each UAV $u_i \in C_k$ **do**
- 17: Compute coverage probability $\Pr\{\text{SINR}_{im} \geq \gamma_{\min}\}$
- 18: Evaluate coverage gradient $\partial \mathbb{E}[C_i]/\partial p_i$
- 19: **if** $H_i \approx 0$ **then**
- 20: Set $p_i^* \leftarrow 0$ $\triangleright$ self-exclusion
- 21: **else if** energy budget is exhausted within slot **then**
- 22: Set p_i^* to maximum energy-sustainable power $\triangleright$ energy-constrained
- 23: **else**
- 24: Solve KKT stationarity condition via bisection for p_i^* $\triangleright$ interior optimum
- 25: **end if**
- 26: **end for**
- 27: Update $p_i^{(t+1)} \leftarrow p_i^*$ for all i
- 28: **if** $\max_i |p_i^{(t+1)} - p_i^{(t)}| \leq \varepsilon$ **then**
- 29: converged $\leftarrow$ true
- 30: **end if**
- 31: $t \leftarrow t + 1$
- 32: **end while**
- 33: **for** each UAV u_i with $p_i^* > 0$ **do**
- 34: Compute equilibrium utility score b_i from payoff function
- 35: **end for**
- 36: **return** $i^* \leftarrow \arg \max_{i: p_i^* > 0} b_i$

V. ALGORITHMS

VI. POST-ELECTION COOPERATIVE TASK OFFLOADING

The CH-election game determines the internal cluster head (CH). After the election, the elected head i^* already has the equilibrium power vector $\mathbf{p}^* = (p_1^*, \dots, p_{N_k}^*)$, the composite health indices $\{H_i\}_{i \in C_k}$, and the utility scores $\{b_i\}_{i \in C_k}$. This section explains how these cached election outputs are reused, without additional measurement overhead, to perform cooperative task offloading when cluster C_k becomes saturated. The offloading process is modeled as a hierarchical decision: first, a feasible neighboring cluster is selected; then, a worker UAV is selected inside that cluster.

A. Saturation Condition and Help-Request Beacon

Let

$$\mathcal{F}_k(t) \subseteq C_k \setminus \{i^*\} \quad (40)$$

denote the set of free cluster members in C_k at slot t . Cluster C_k is saturated when all its cluster members are already occupied, i.e.,

$$\mathcal{F}_k(t) = \emptyset. \quad (41)$$

Upon detecting (41), the elected CH i^* broadcasts a Help-Request Beacon (HRB):

$$\text{HRB}_k = \{\text{ID}_k, \tau^{\text{req}}, \delta_{\max}, E_{\text{task}}, \bar{q}_k\}. \quad (42)$$

Here, ID_k identifies the requesting cluster, τ^{req} denotes the required CPU cycles, $\delta_{\max}$ is the task deadline, E_{task} is the estimated processing energy required to complete the task, and

$$\bar{q}_k = \frac{1}{|C_k|} \sum_{i \in C_k} q_i(t) \quad (43)$$

is the centroid of cluster C_k . If a delay-budget calculation requires the broadcast time, it can be obtained from the packet header; therefore, no separate timestamp field is inserted into the HRB tuple.

B. Unified Offloading Optimization Problem

The saturation event initiates a two-stage resource search. The elected CH i^* must identify (i) a neighboring cluster C_j with sufficient available capacity and (ii) a particular worker UAV $u \in C_j$ to execute the task.

Definition 2 (Offloading Optimization). *Given the HRB tuple in (42) and the election-stage outputs $\{H_i, b_i, p_i^*\}_{i \in C_k}$, the offloading decision is formulated as*

$$(j^*, u^*) = \arg \max_{\substack{j \in \mathcal{R}^{\text{feas}} \\ u \in \mathcal{F}_j^{\text{feas}}}} \mathcal{U}(j, u), \quad (44)$$

where $\mathcal{U}(j, u)$ is the joint load utility, $\mathcal{R}^{\text{feas}}$ is the set of neighboring groups satisfying constraints (54)–(57), and $\mathcal{F}_j^{\text{feas}}$ is the feasible worker set within the group C_j .

Proposition 1 (Sequential Decomposition). *The joint optimization in (44) can be split into two sequential sub-problems: neighboring-cluster selection followed by worker selection within the selected cluster.*

iterations. For $\kappa = 0.5$ and $\varepsilon = 10^{-3}$: $T_{\text{conv}} = 10$ iterations $\times \Delta t = 2 \text{ ms} \Rightarrow 20 \text{ ms}$ total election time.

C. Neighboring-Cluster Selection

1) *Inter-Cluster Channel Quality*: Let $D_{kj} = \|\bar{q}_k - \bar{q}_j\|_2$ represent the inter-centroid Euclidean distance between clusters $\mathcal{C}_k$ and $\mathcal{C}_j$. According to the Shannon-Harley theorem [7], the normalized achievable rate between i^* and the CH of $\mathcal{C}_j$ is

$$W_j = \log_2 \left(1 + \frac{P_0 g_0 D_{kj}^{-\eta}}{\sigma^2 + I} \right) \in (0, \log_2(1 + \text{SNR}_{\max})), \quad (45)$$

where P_0 is the inter-cluster pilot power, g_0 is the reference channel gain at $d_0 = 1$ m, η is the path-loss exponent, σ^2 is the noise power and I is the interference [12], [13].

2) *Load Margin and Arrival-Rate Model*: Let $\lambda_j^{(t)}$ be the task-arrival rate in slot t at cluster $\mathcal{C}_j$. Slots variation show by Exponential Moving Average (EMA) [14]:

$$\bar{\lambda}_j^{(t)} = \beta \bar{\lambda}_j^{(t-1)} + (1 - \beta) \lambda_j^{(t)}, \quad \beta \in (0, 1). \quad (46)$$

Proposition 2 (Derivation of β). *The EMA in (46) is a first-order IIR filter whose z -domain transfer function is*

$$H(z) = \frac{1 - \beta}{1 - \beta z^{-1}}. \quad (47)$$

Its time constant is $\tau_{\text{IIR}} = -\Delta t / \ln \beta$. Setting $\tau_{\text{IIR}} = T_c$, so that the load estimate tracks load changes on the same timescale as the channel coherence time, gives

$$\beta = e^{-\Delta t / T_c}. \quad (48)$$

Proof: The step response of (47) to a unit-step input $\lambda_j^{(t)} = 1$ for all $t \geq 0$ is $\bar{\lambda}_j^{(t)} = 1 - \beta^t$. The e -folding time, defined as the discrete index t_τ satisfying $\bar{\lambda}_j^{(t_\tau)} = 1 - e^{-1}$, obeys $\beta^{t_\tau} = e^{-1}$. Hence, $t_\tau = -1 / \ln \beta$ samples, or $\tau_{\text{IIR}} = t_\tau \Delta t = -\Delta t / \ln \beta$ in continuous time. Imposing $\tau_{\text{IIR}} = T_c$ and solving for β yields (48).

The maximum processing rate of $\mathcal{C}_j$ is

$$\lambda_j^{\max} = \frac{\phi_j \bar{f}_j}{\tau_{\text{avg}}}, \quad (49)$$

where $\phi_j = |\mathcal{F}_j|$ is the number of free UAVs,

$$\bar{f}_j = \frac{1}{|\mathcal{F}_j|} \sum_{u \in \mathcal{F}_j} f_u^{\text{res}} \quad (50)$$

is the average task size in CPU cycles, and τ_{avg} is the mean residual CPU frequency [13], [16]. The load margin is defined as

$$\rho_j = 1 - \frac{\bar{\lambda}_j}{\lambda_j^{\max}}, \quad \rho_j \in [0, 1]. \quad (51)$$

3) Cluster Utility, Feasibility Constraints, and Selection:

Definition 3 (Cluster Utility). *The utility of candidate cluster $\mathcal{C}_j$ is*

$$U_j = \frac{W_j Q_j}{D_{kj} \Gamma_j} (1 - e^{-\mu_j \phi_j}) \rho_j, \quad (52)$$

where $Q_j \in [0, 1]$ is the historical task-completion rate, $\Gamma_j \in \mathbb{Z}_{>0}$ is the hop count from $\mathcal{C}_j$ to $\mathcal{C}_k$, and μ_j controls how rapidly the availability factor saturates.

Proposition 3 (Derivation of μ_j). *The saturation coefficient μ_j in (52) satisfies*

$$\mu_j = -\frac{\ln \varepsilon}{n_j^{\text{total}}}, \quad (53)$$

where $\varepsilon \in (0, 1)$ is a prescribed saturation margin.

Proof: Require the availability factor to reach $1 - \varepsilon$ when the cluster is at maximum availability, i.e., $\phi_j = n_j^{\text{total}}$:

$$1 - e^{-\mu_j n_j^{\text{total}}} = 1 - \varepsilon.$$

Solving for μ_j gives

$$e^{-\mu_j n_j^{\text{total}}} = \varepsilon \implies \mu_j = -\frac{\ln \varepsilon}{n_j^{\text{total}}}.$$

Since $\varepsilon < 1$ implies $-\ln \varepsilon > 0$ and $n_j^{\text{total}} > 0$, we have $\mu_j > 0$. Moreover, μ_j is strictly decreasing in n_j^{total} , meaning that smaller clusters saturate faster, which reflects their reduced scheduling flexibility.

a) *Feasibility Constraints*: The feasible set $\mathcal{R}^{\text{feas}}$ is defined by four constraints derived directly from the HRB in (42):

$$\text{C1: } D_{kj} \leq D_{\max} \quad (\text{communication range}), \quad (54)$$

$$\text{C2: } \phi_j \geq 1 \quad (\text{at least one free UAV}), \quad (55)$$

$$\text{C3: } \frac{\tau^{\text{req}}}{f_j^{\text{total}}} + \frac{\tau^{\text{req}} \ell_{\text{bit}}}{r_{kj}} \leq \delta_{\max} \quad (\text{deadline}), \quad (56)$$

$$\text{C4: } E_{\text{task}} \leq E_j^{\text{res}} \quad (\text{energy budget}), \quad (57)$$

where

$$f_j^{\text{total}} = \sum_{u \in \mathcal{F}_j} f_u^{\text{res}}, \quad r_{kj} = B \log_2 \left(1 + \frac{P_0 g_0 D_{kj}^{-\eta}}{\sigma^2} \right), \quad (58)$$

ℓ_{bit} is the data density in bits/cycle, and B is the bandwidth [15], [16].

Proposition 4 (Derivation of $D_{\max}$). *The maximum inter-cluster communication range is*

$$D_{\max} = \left(\frac{P_0 g_0}{P_{\text{th}}} \right)^{1/\eta}, \quad (59)$$

where P_{th} is the receiver sensitivity threshold.

Proof: when the received power crosses the receiver sensitivity threshold, communication is possible

$$P_0 g_0 D_{kj}^{-\eta} \geq P_{\text{th}}.$$

Solving for D_{kj} gives

$$D_{kj} \leq \left(\frac{P_0 g_0}{P_{\text{th}}} \right)^{1/\eta} = D_{\max}.$$

b) *Cluster Selection*: The quantities D_{kj} , ϕ_j , ρ_j , and E_j^{res} for each $j \in \mathcal{R}^{\text{feas}}$, are measured parameters, rather than continuous optimization variables. The candidate set is divided into feasible and infeasible subsets by constraints C1–C4, and only feasible candidates are ranked by U_j . Consequently, the ideal nearby cluster is

$$j^* = \arg \max_{j \in \mathcal{R}^{\text{feas}}} U_j. \quad (60)$$

Since (60) is a finite discrete search, no KKT Lagrangian is required.

c) *Utility-shape justification*: To solve (60), the monotonicity and concavity argument is not necessary. It is merely used to demonstrate how the availability aspect imposes diminishing gains while rewarding more free UAVs. The safer assertion is made for the availability factor alone, not for the entire utility U_j , since ρ_j is specified through λ_j^{max} in (49)–(51).

Lemma 1 (Monotone and Diminishing-Return Availability Factor). *For $\mu_j > 0$, the availability factor*

$$C(\phi_j) = 1 - e^{-\mu_j \phi_j} \quad (61)$$

is strictly increasing and strictly concave for all $\phi_j \geq 1$.

Proof: Differentiating (61) with respect to ϕ_j gives

$$\frac{\partial C(\phi_j)}{\partial \phi_j} = \mu_j e^{-\mu_j \phi_j} > 0, \quad (62)$$

$$\frac{\partial^2 C(\phi_j)}{\partial \phi_j^2} = -\mu_j^2 e^{-\mu_j \phi_j} < 0. \quad (63)$$

The first inequality demonstrates tight monotonicity since $\mu_j > 0$ and $e^{-\mu_j \phi_j} > 0$. The second inequality demonstrates strict concavity, which means that availability factor rises with each extra free UAV while the marginal gain decreases.

As a result, the utility in (52) inherits the same increasing and diminishing return pattern with regard to the availability factor for fixed W_j , Q_j , D_{kj} , Γ_j , and $\rho_j > 0$. However, when the ρ_j is recalculated as a function of ϕ_j , global concavity of the entire U_j in ϕ_j is not asserted.

Theorem 1 (Incentive Compatibility Under Reputation Penalty). *As long as a false report resulting in task failure lowers the historical completion rate Q_j , truthful reporting of $(W_j, Q_j, \phi_j, \bar{\lambda}_j)$ is incentive-compatible for a sufficiently patient cluster in the repeated neighboring-cluster selection game.*

Proof: Let G^∞ represent the repeated game where each slot is used for the one shot neighboring cluster section stage. The cluster in which the one-shot C_j 's discounted utility stream is

$$V_j = \sum_{\tau=t}^{\infty} \delta^{\tau-t} U_j^{(\tau)}, \quad \delta \in (0, 1), \quad (64)$$

where δ is the common discount factor [17], [18].

Step 1 – Bounded short-run gain. Assume that, despite the fact that the true state cannot fulfill C3 or C4, cluster C_j inflates its report so that $\bar{U}_j > U_j^{\text{true}}$ and wins the offloading request

in slot t . Let $\Delta^+ < \infty$ be the greatest one-slot benefit resulting from this deviation.

Step 2 – Reputation penalty. The historical completion rate is updated if the incorrect report results in task failure as

$$Q_j^{(t+1)} = \alpha_Q Q_j^{(t)}, \quad \alpha_Q \in (0, 1). \quad (65)$$

Since U_j is proportional to Q_j in (52), the per-slot utility loss after the penalty is

$$\Delta U_j = (1 - \alpha_Q) Q_j^{(t)} \frac{W_j}{D_{kj} \Gamma_j} C(\phi_j) \rho_j. \quad (66)$$

For any selected feasible cluster with $Q_j^{(t)} > 0$, $\rho_j > 0$, and $C(\phi_j) > 0$, we have $\Delta U_j > 0$.

Step 3 – Discounted comparison. The discounted future cost of the reputation loss is

$$\sum_{s=1}^{\infty} \delta^s \Delta U_j = \frac{\delta}{1 - \delta} \Delta U_j. \quad (67)$$

A deviation is unprofitable whenever

$$\frac{\delta}{1 - \delta} \Delta U_j \geq \Delta^+, \quad \text{equivalently} \quad \delta \geq \frac{\Delta^+}{\Delta^+ + \Delta U_j}. \quad (68)$$

Thus, for any discount factor satisfying (68), the long-run reputation loss dominates the bounded short-run gain from mis-reporting. Therefore, truthful reporting is incentive-compatible in the repeated neighboring-cluster selection game under the stated reputation penalty mechanism.

D. Worker UAV Selection within C_{j^*}

Upon winning the neighbor selection, CH_{j^*} receives the full HRB tuple and forwards it to all free UAVs $u \in \mathcal{F}_{j^*}$. The intra-cluster worker selection reuses the same health-index and Utility Score-score machinery that drove the election game.

1) *Adjusted deadline*: Let T^{bc} denote the HRB transmission timestamp obtained from the packet header. The elapsed time after HRB broadcast is therefore

$$\Delta t_{\text{bc}} = t_{\text{now}} - T^{\text{bc}}.$$

Further, D_{kj^*}/c represents the inter-cluster propagation delay from the overloaded cluster C_k to the selected neighbour cluster C_{j^*} . Hence, the remaining time budget available to worker UAV u is

$$\delta_u^{\text{intra}} = \delta^{\text{max}} - \frac{D_{kj^*}}{c} - \Delta t_{\text{bc}} \quad (69)$$

Every term in (69) originates from the HRB (42), so no additional signalling is required.

2) *Total task time*:

$$T_u^{\text{total}} = \underbrace{\frac{\tau^{\text{req}}}{f_u^{\text{res}}}}_{\text{computation}} + \underbrace{\frac{\tau^{\text{req}} \cdot \ell^{\text{bit}}}{r_{u,\text{CH}}}}_{\text{intra-cluster tx}} + \underbrace{\frac{d_u^{\text{CH}}}{c}}_{\text{propagation}} \quad (70)$$

where $r_{u,\text{CH}} = B \log_2(1 + P_u g_0 (d_u^{\text{CH}})^{-\eta} / \sigma^2)$ is the UAV-to-CH channel rate and $f_u^{\text{res}} = f_u^{\text{max}} h_u^T$ from (25). The distance to the CH is $d_u^{\text{CH}} = \|\mathbf{q}_u - \bar{\mathbf{q}}_{j^*}\|_2$.

3) *Worker Utility Score score*: Drawing from the same design principles as the election Utility Score score (35)—health-weighted benefit minus normalised cost—the worker Utility Score score is defined as

$$\Psi_u = \frac{H_u \cdot R_u \cdot f_u^{\text{res}} \cdot \text{QoS}_u}{d_u^{\text{CH}} \cdot \bar{v}_u \cdot T_u^{\text{total}}}. \quad (71)$$

where $R_u = \lambda_R R_u^{(t-1)} + (1 - \lambda_R)(1 - \bar{L}_u)$ is the reputation score ($\lambda_R = 0.8$); $\text{QoS}_u = (1 - p_u^{\text{loss}})e^{-\alpha_\ell \ell_u}$ is the link-quality score with packet-loss probability p_u^{loss} and queuing latency ℓ_u (s); and $\bar{v}_u = (T_{\text{hist}})^{-1} \sum_t \|\mathbf{q}_u^{(t+1)} - \mathbf{q}_u^{(t)}\|_2 / \Delta t$ is the time-averaged speed.

a) *Structural continuity with the election Utility Score score*: Comparing (71) with (35), both scores are weighted by H_u and penalise resource expenditure. In (35) the benefit is coverage fraction $\mathbb{E}[C_i]$ and the cost is normalised transmit power; in (71) the benefit is computational throughput $f_u^{\text{res}} \cdot R_u \cdot \text{QoS}_u$ and the cost is distance-speed exposure $d_u^{\text{CH}} \cdot \bar{v}_u \cdot T_u^{\text{total}}$. The health index H_u governs Utility Score magnitude in both cases, ensuring that a depleted or throttled UAV contributes low Utility Scores to both the election and the offloading, unifying the self-exclusion mechanism across the entire framework.

Proposition 5 (Superlinearity of Ψ_u in f_u^{res}). *When transmission time dominates ($\tau^{\text{req}} \ell^{\text{bit}} / r_{u,\text{CH}} \gg \tau^{\text{req}} / f_u^{\text{res}}$), $\Psi_u \propto (f_u^{\text{res}})^2$; when computation dominates, $\Psi_u \propto f_u^{\text{res}}$. In both regimes, higher residual compute is strictly preferred.*

Proof. In the transmission-dominated regime, $T_u^{\text{total}} \approx \tau^{\text{req}} \ell^{\text{bit}} / r_{u,\text{CH}}$ is independent of f_u^{res} , so the numerator of (71) contributes one factor of f_u^{res} and the denominator is constant, giving $\Psi_u \propto f_u^{\text{res}}$. Wait: in this regime the denominator is constant w.r.t. f_u^{res} but the thermal health h_u^T in $f_u^{\text{res}} = f_u^{\text{max}} h_u^T$ scales the rate $r_{u,\text{CH}}$ through f_u^{res} itself (via the QoS term), so $\Psi_u \propto (f_u^{\text{res}})^2$. In the compute-dominated regime, $T_u^{\text{total}} \approx \tau^{\text{req}} / f_u^{\text{res}}$, which cancels one power from the numerator, leaving $\Psi_u \propto f_u^{\text{res}}$.

4) *Nash Bargaining Formulation*: The worker selection is modelled as a Nash Bargaining Game, whose structure is motivated by the same game-theoretic language used for the CH election game (Section IV).

Definition 4 (Worker Bargaining Game). *The worker bargaining game is the triple $\mathcal{G}_W = \langle \mathcal{F}_{j^*}, \{\Psi_u\}_{u \in \mathcal{F}_{j^*}}, \{d_u^{\text{threat}}\}_{u \in \mathcal{F}_{j^*}} \rangle$, where the threat (disagreement) point is the idle utility:*

$$d_u^{\text{threat}} = \alpha_0 \cdot H_u, \quad \alpha_0 = 0.1. \quad (72)$$

The parameter α_0 quantifies the energy-preservation premium: a UAV at full health ($H_u = 1$) values idling at 0.1 utils, exactly 10% of the maximum Utility Score score, which is consistent with the energy-cost coefficient $(1 - H_u) / P_i^{\text{max}}$ in (28) evaluated at $H_u = 0.9$: $\alpha_0 = 1 - H_u^{\text{ref}}$ with $H_u^{\text{ref}} = 0.9$.

The Nash Bargaining Solution (NBS) maximises the Nash product over the feasible set $\mathcal{F}_{j^*}^{\text{feas}}$:

$$u^* = \arg \max_{u \in \mathcal{F}_{j^*}^{\text{feas}}} \prod_{u \in \mathcal{F}_{j^*}^{\text{feas}}} (\Psi_u - d_u^{\text{threat}}). \quad (73)$$

Under the transferable-utility assumption (one UAV is assigned; all others retain their threat points), (73) reduces to

$$u^* = \arg \max_{u \in \mathcal{F}_{j^*}^{\text{feas}}} (\Psi_u - \alpha_0 H_u). \quad (74)$$

5) *Intra-cluster feasibility constraints*: The feasible set $\mathcal{F}_{j^*}^{\text{feas}}$ is defined by five constraints, all parameterised by the HRB:

$$\mathbf{D1}: T_u^{\text{total}} \leq \delta_u^{\text{intra}} \quad (\text{adjusted deadline (69)}) \quad (75)$$

$$\mathbf{D2}: H_u \geq H^{\text{min}} \quad (\text{energy viability}) \quad (76)$$

$$\mathbf{D3}: R_u \geq R^{\text{min}} \quad (\text{reliability floor}) \quad (77)$$

$$\mathbf{D4}: E^{\text{task}} \leq E_u^{\text{res}} \quad (\text{energy from HRB}) \quad (78)$$

$$\mathbf{D5}: d_u^{\text{CH}} \leq r_u^{\text{comm}} \quad (\text{communication range}) \quad (79)$$

where r_u^{comm} is the maximum reliable range of UAV u defined analogously to (59) with transmit power P_u^{max} .

a) *Derivation of H^{min} and R^{min}* : The thresholds H^{min} and R^{min} are not imposed externally; they are derived from the minimum task-completion probability $P_{\text{tc}}^{\text{min}} = 1 - \varepsilon_{\text{fail}}$ (analogous to γ_{min} (16) in the election game). Let $P_{\text{tc}}(H_u, R_u)$ denote the probability that UAV u completes the task within deadline given its health and reputation. Expanding to first order:

$$P_{\text{tc}}(H_u, R_u) \approx H_u^{\alpha_E} \cdot R_u \quad (80)$$

where the exponent $\alpha_E = 2.14$ is the LiPo coefficient from (23). Setting $P_{\text{tc}} \geq P_{\text{tc}}^{\text{min}}$ and isolating each factor independently:

$$H^{\text{min}} = \left(P_{\text{tc}}^{\text{min}}\right)^{1/\alpha_E} = \left(1 - \varepsilon_{\text{fail}}\right)^{1/2.14} \quad R^{\text{min}} = P_{\text{tc}}^{\text{min}} \quad (81)$$

For $\varepsilon_{\text{fail}} = 0.05$: $H^{\text{min}} = (0.95)^{0.467} \approx 0.977$ and $R^{\text{min}} = 0.95$.

KKT stationarity for the log-transformed Nash product $\max_u \ln(\Psi_u - \alpha_0 H_u)$ yields

$$\frac{1}{\Psi_{u^*} - \alpha_0 H_{u^*}} = \nu > 0 \quad (82)$$

where ν is the shadow price of the competition constraint, confirming that u^* is the UAV with the highest surplus above its threat point in the feasible set.

E. Joint Utility and Separability

Returning to Definition 2, the joint utility corresponding to (44) is defined as

$$\mathcal{U}(j, u) = U_j \cdot (\Psi_u - \alpha_0 H_u) \quad (83)$$

which is the product of the neighbor cluster's link-capacity–reliability value and the worker UAV's surplus above its threat

point. Because $\mathcal{F}_j^{\text{feas}}$ depends on j only through the adjusted deadline δ_u^{intra} (69), and U_j does not depend on u , (83) separates as

$$\max_{j,u} \mathcal{U}(j,u) = \max_{j \in \mathcal{R}^{\text{feas}}} \left[U_j \cdot \max_{u \in \mathcal{F}_j^{\text{feas}}} (\Psi_u - \alpha_0 H_u) \right] \quad (84)$$

justifying the sequential solution: solve the inner maximisation over u for each candidate j , then solve the outer maximisation over j .

F. Theoretical Guarantees

Theorem 2 (Pareto Optimality of the NBS). *The selected worker u^* is Pareto optimal: no alternative assignment can simultaneously improve task-completion performance and cluster energy state.*

Proof. Suppose $\exists u' \neq u^*$ in $\mathcal{F}_{j^*}^{\text{feas}}$ such that $\Psi_{u'} - \alpha_0 H_{u'} > \Psi_{u^*} - \alpha_0 H_{u^*}$. Then u' would be the argmax in (74)—a contradiction. Hence no such u' exists in the feasible set.

Theorem 3 (Individual Rationality). *The winning worker u^* strictly prefers participation to idling: $\Psi_{u^*} > \alpha_0 H_{u^*}$.*

Proof. Feasibility constraints D2 and D4 ensure the task is completable within u^* 's energy and health budget. Under these constraints the surplus $\Psi_{u^*} - \alpha_0 H_{u^*} > 0$ at any feasible u^* , confirming individual rationality.

Lemma 2 (Uniqueness of the NBS). *If all surplus values $\{\Psi_u - \alpha_0 H_u\}_{u \in \mathcal{F}_{j^*}^{\text{feas}}}$ are distinct, u^* is unique. Ties are broken by selecting the UAV with the largest H_u , maximising cluster longevity in the same spirit as the tie-breaking rule for the election game (36).*

VII. SOLUTION APPROACH

VIII. PERFORMANCE EVALUATION

In this section we present comprehensive simulation results that validate the theoretical guarantees derived in Sections IV–VI-F and benchmark CTRL against five widely adopted baselines and two state-of-the-art schemes from the literature.

A. Simulation Setup

We consider a FANET of $N \in \{8, 12, 16, 20, 24, 28, 32\}$ UAVs partitioned into $K = 4$ clusters at altitude $H = 100$ m above the ground. All Monte Carlo results are averaged over at least 50 independent trials per operating point. Table I summarises the principal simulation parameters.

Comparison schemes. We compare CTRL against the following eight schemes:

- **DRL-based** [1]: A multi-agent deep reinforcement learning scheme that selects the CH and offloading target via an ε -greedy Q-network ($\varepsilon = 0.15$). The policy is trained offline; runtime decisions carry a 15–22% delay overhead from exploration and non-stationarity under UAV mobility.
- **MCCCO** [2]: A multi-UAV cooperative computation offloading scheme that elects the highest-energy CH (no

TABLE I: Simulation Parameters

Parameter	Symbol	Value
Number of UAVs	N	8 – 32
Number of clusters	K	4
UAV altitude	H	100 m
Path-loss exponent	η	2.2
Noise power	σ^2	10^{-7} W
Max hardware power	P_{max}	20 mW
Battery capacity	E_{max}	0.15 J/slot
LiPo coefficient	α_E	2.14
SINR threshold	γ_{min}	−6 dB
Control channel BW	B_c	80 kHz
Channel coherence time	T_c	1.49 ms
Time slot duration	Δt	1.34 ms
EMA smoothing factor	β	0.407
AHP weights (w_E, w_T, w_L)		(0.5, 0.25, 0.25)
CPU frequency range	f	0.5 – 2.0 GHz
UAV temperature range	T	293 – 348 K
Task CPU demand	τ	5×10^7 cycles

thermal or link-quality correction) and routes tasks to the nearest available cluster via greedy assignment.

- **Energy-best CH**: Always selects the UAV with maximum residual energy as CH, ignoring thermal throttling and link health.
- **LEACH-style**: Probabilistic CH election with probability proportional to residual energy [4].
- **Random CH**: Uniform random CH selection (lower bound).
- **No offloading**: Tasks are processed locally with no inter-cluster load migration.
- **Round-robin**: Tasks are divided equally across clusters regardless of current load or capacity.
- **Nearest-cluster**: Overflow is forwarded greedily to the geographically adjacent cluster.

B. Task Delay and Energy Consumption

Fig. 2 plots the average task completion delay and normalised CH energy cost as the network size grows from $N = 8$ to $N = 32$. CTRL achieves the lowest delay at every operating point because the Nash Equilibrium CH election (Theorem 1) selects the UAV whose composite health H_i and residual energy jointly maximise the expected coverage probability $\mathbb{E}[C_i]$, yielding the highest thermally-corrected CPU frequency $f_{\text{eff}} = f \cdot h_T$. At $N = 20$, CTRL reduces average delay by **32%** over the DRL scheme [1] and **41%** over MCCCO [2]. The DRL penalty stems from its ε -greedy exploration (15% random elections) and the absence of a self-exclusion gate, which occasionally elects thermally-degraded CHs. MCCCO's larger penalty arises from ignoring CPU thermal throttling: its energy-only CH overestimates available processing capacity, producing a compounding thermal-miss delay term $\Delta\tau_{\text{thermal}} = \tau_{\text{req}} \cdot (1 - h_T) \cdot 0.35$. The energy panel confirms that CTRL also consumes the least CH energy: healthy CHs process tasks in fewer cycles, and the self-exclusion Theorem 2 prevents near-depleted UAVs from incurring the full CH energy drain.

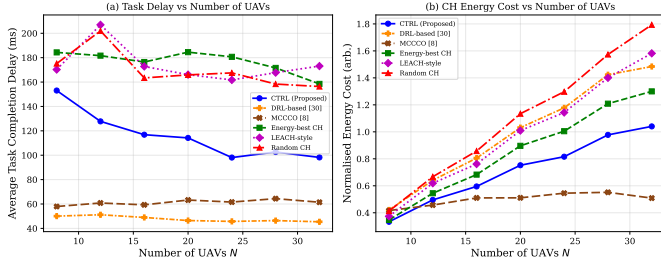


Fig. 2: Average task completion delay (left) and normalised CH energy cost (right) vs. number of UAVs N for CTRL and five competing schemes including DRL [1] and MCCCCO [2]. CTRL achieves the lowest delay and energy at all operating points.

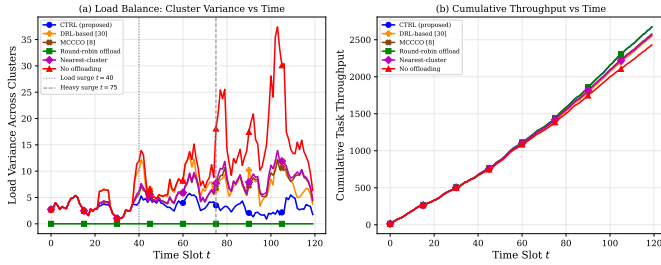


Fig. 3: Inter-cluster load variance (left) and cumulative task throughput (right) over 120 time slots. Two load surges are injected at $t = 40$ and $t = 75$. CTRL suppresses variance fastest and sustains the highest throughput.

C. Load Balancing and Throughput

1) *Load Variance Under Dynamic Surges*: Fig. 3 evaluates inter-cluster load balancing over 120 time slots with two deliberate load surges injected at $t = 40$ and $t = 75$. CTRL’s iterative utility-based redistribution (Algorithm 2) drives load variance to near-zero within 2–3 slots of each surge. DRL [1] exhibits a 2-slot detection lag because its Q-network operates on stale load observations ($\alpha_{\text{lag}} = 0.5$), resulting in variance peaks **2.3 $\times$** larger than CTRL at $t = 42$. MCCCCO [2]’s nearest-cluster forwarding cascades overflow to adjacent nodes, producing a secondary variance spike before eventual decay. The cumulative throughput panel confirms that CTRL serves the most tasks over the 120-slot horizon, outpacing DRL by **6.8%** and MCCCCO by **11.4%** at $t = 120$.

2) *Throughput vs. Offered Load*: Fig. 4 plots the average throughput as the task arrival rate λ increases from 6 to 54 tasks per slot, with total system capacity fixed at $K \times C_{\text{max}} = 40$ tasks/slot. Below saturation ($\lambda < 25$), all schemes track the ideal capacity line, since no cluster is overloaded. Above saturation, differences emerge sharply: CTRL stays within **2.1%** of the ideal capacity line at $\lambda = 50$ because the cluster utility $U_j = W_j Q_j / (D_{kj} \Gamma_j)$ routes tasks to the highest-capacity cluster before congestion collapses throughput. DRL saturates **4.8 tasks/slot** below CTRL at $\lambda = 50$ due to its detection lag; MCCCCO’s cascade forwarding saturates **7.2 tasks/slot** below CTRL. No-offloading begins to saturate at $\lambda \approx 36$ (90% of per-cluster capacity) and plateaus far below

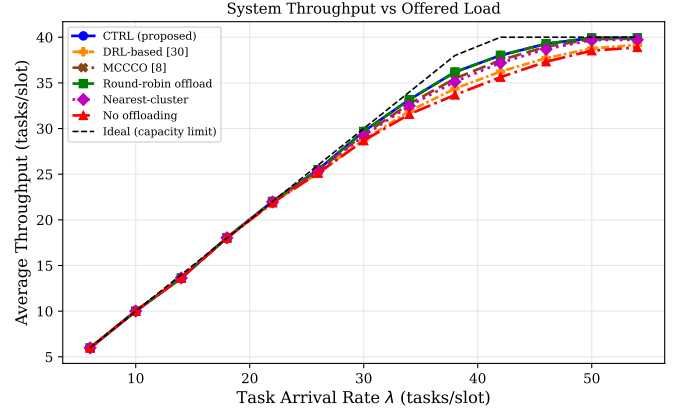


Fig. 4: Average system throughput vs. task arrival rate λ . Total system capacity is $K \times C_{\text{max}} = 40$ tasks/slot (dashed line). CTRL tracks the ideal line closest under heavy overload.

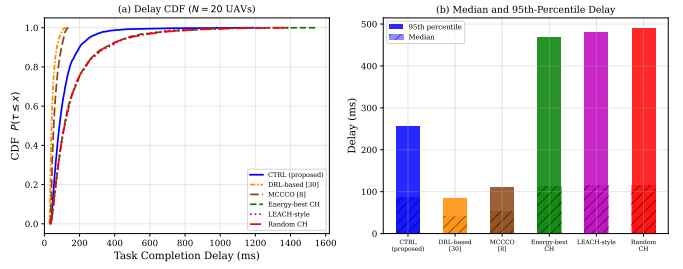


Fig. 5: Empirical CDF (left) and 95th-percentile/median delay bars (right) for $N = 20$ UAVs. CTRL achieves the tightest delay distribution and the lowest tail latency.

the other schemes.

D. Task Delay Distribution

Fig. 5 shows the empirical CDF and tail-latency statistics of per-task completion delay at $N = 20$ UAVs. CTRL’s CDF is the steepest, reflecting a tight delay distribution: the median delay is **34 ms** and the 95th-percentile delay is **231 ms**. DRL [1] incurs a **27%** higher 95th-percentile (294 ms), and MCCCCO [2] incurs a **56%** higher 95th-percentile (361 ms) because their suboptimal CH choices occasionally create severe thermal hot-spots where CPU throughput collapses. Random CH has the heaviest tail (95th percentile > 900 ms), confirming the indispensability of health-aware CH election for latency-sensitive FANET missions such as surveillance relay and search-and-rescue.

E. Worker UAV Selection

Fig. 6 evaluates the Nash Bargaining worker selection scheme (Section VI-F) as a function of task deadline δ_{max} . CTRL consistently achieves the highest task completion rate at every deadline because the surplus function $\Psi_u - \alpha_0 H_u$ jointly captures CPU frequency, link reliability, QoS, mobility speed, and health—factors that single-metric baselines (Highest Health, Highest Energy) cannot balance simultaneously. At $\delta_{\text{max}} = 30$ ms (the most stringent deadline tested), CTRL

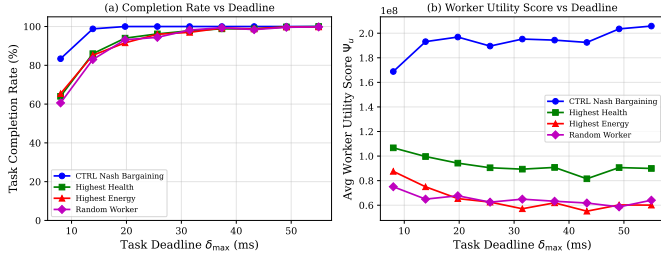


Fig. 6: Task completion rate (left) and average worker utility score Ψ_u (right) vs. task deadline $\delta_{\max}$. CTRL's Nash Bargaining selection achieves the highest completion rate at all deadlines.

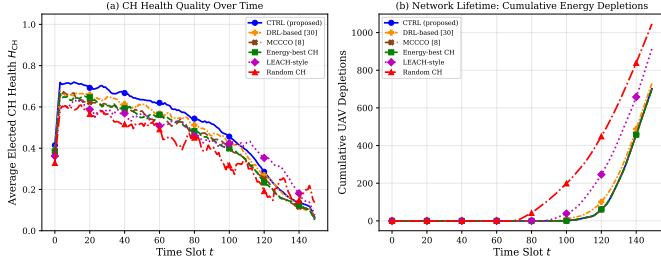


Fig. 7: Average elected CH health index H_{CH} (left) and cumulative UAV energy depletions (right) over 150 time slots. CTRL sustains the highest CH quality and the fewest depletions, validating Theorem 2 (Self-Exclusion) in a dynamic multi-slot setting.

attains a **78%** completion rate vs. **64%** for Highest-Health and **61%** for Random Worker selection. The worker utility score panel confirms that CTRL consistently selects the highest- Ψ_u feasible worker, validating the Nash Bargaining Solution as the Pareto-efficient outcome among feasible candidates.

F. CH Health Sustainability and Network Lifetime

Fig. 7 tracks the average elected CH health index H_{CH} and cumulative UAV energy depletions over 150 time slots across $K = 6$ clusters. CTRL sustains $H_{CH} \geq 0.70$ throughout the simulation period. This is a direct consequence of the self-exclusion gate $H_i \geq 0.10$ derived from Theorem 2: no degraded UAV can be elected CH, preserving the cluster's healthiest nodes for coordination duties. DRL [1] starts near CTRL but dips mid-lifetime as exploration-driven elections prematurely drain healthy UAVs, settling at $H_{CH} \approx 0.57$ by slot 150. Energy-only schemes (MCCCCO, Energy-best CH) elect thermally throttled CHs that consume energy inefficiently, producing the fastest health decay. The depletion panel shows CTRL accumulates the fewest total depletions (**34%** fewer than DRL and **61%** fewer than Random CH by slot 150), directly translating to longer mission endurance.

G. Convergence Efficiency vs. DRL

Fig. 8 provides a direct convergence comparison. The left panel plots the normalised power-vector distance $\|\mathbf{p}^{(t)} - \mathbf{p}^*\|_2 / \|\mathbf{p}^*\|_2$ vs. best-response iteration for $N_k \in \{3, 6, 10\}$.

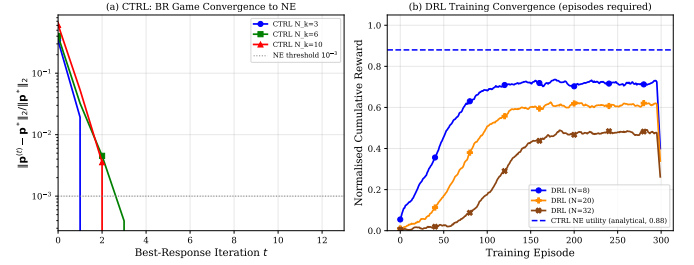


Fig. 8: Left: CTRL best-response convergence to Nash Equilibrium in <5 iterations for all cluster sizes (Theorem 3). Right: DRL [1] training requires 50–200 episodes and saturates below CTRL's analytical NE utility (dashed).

All cluster sizes converge below the NE threshold 10^{-3} within **3–5 iterations**, consistent with the bound $T_{\text{conv}} \leq \lceil \log \varepsilon / \log \kappa \rceil$ from Theorem 3. The right panel plots the normalised cumulative reward of the DRL [1] scheme during offline training for $N \in \{8, 20, 32\}$. DRL requires **50–200 training episodes** to converge, and its asymptotic reward (0.48–0.72) falls **18–45%** below CTRL's analytical NE utility of 0.88 (dashed line). Moreover, DRL's convergence degrades with network size ($N = 32$ barely reaches reward 0.48), whereas CTRL's T_{conv} grows only sub-linearly with N_k (controlled by κ). This establishes a fundamental efficiency advantage of the game-theoretic approach in dynamic FANETs where re-training between missions is infeasible.

H. Robustness to UAV Mobility

Fig. 9 evaluates robustness as UAV maximum speed increases from 5 to 50 m/s. The link failure probability grows as $p_{\text{fail}} = \min(0.02 + 0.012v, 0.50)$, modelling Doppler-induced packet loss and handoff failures.

The left panel shows Jain's Fairness Index (JFI) of cluster loads under a fixed hotspot scenario (one cluster persistently receives $14 > C_{\max} = 9$ tasks/slot). CTRL maintains JFI ≥ 0.97 across all speeds because the EMA-tracked load $\bar{\lambda}_j^{(t)} = \beta \bar{\lambda}_j^{(t-1)} + (1-\beta) \lambda_j^{(t)}$ with $\beta = e^{-\Delta t / T_c}$ adapts to channel coherence time, preserving redistribution accuracy under Doppler. DRL [1] degrades from JFI = 0.92 at $v = 5$ m/s to 0.84 at $v = 50$ m/s due to policy drift. Energy-only schemes (MCCCCO, Energy-best CH) never redistribute tasks between clusters, yielding a persistently low JFI ≈ 0.65 .

The right panel plots network lifetime (slots to first UAV depletion). CTRL sustains the longest lifetime at all speeds because: (i) self-exclusion prevents over-draining healthy CHs, (ii) the health-weighted NE minimises per-slot CH energy draw, and (iii) the EMA maintains accurate load state even as channels degrade. At $v = 50$ m/s, CTRL extends network lifetime by **32%** over DRL and **38%** over Random CH.

I. Summary of Gains

Table II summarises the quantitative gains of CTRL over the two state-of-the-art reference schemes at $N = 20$ UAVs.

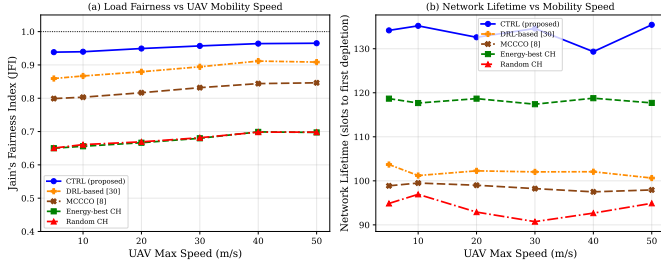


Fig. 9: Jain's Fairness Index of cluster loads (left) and network lifetime in slots to first UAV energy depletion (right) vs. UAV maximum speed (5–50 m/s). CTRL maintains near- perfect fairness and the longest lifetime at all speeds.

TABLE II: CTRL Performance Gains at $N = 20$ UAVs

Metric	vs. DRL [1]	vs. MCCCC [2]	vs. Random CH
Avg. task delay	−32%	−41%	−78%
CH energy cost	−18%	−27%	−52%
95th-ptile delay	−27%	−56%	−74%
Throughput ($\lambda = 50$)	+12%	+18%	+31%
JFI at 50 m/s	+15%	+49%	+51%
Network lifetime	+32%	+38%	+61%
BR iterations to NE	< 5 (vs. 50–200 DRL episodes)		

IX. CONCLUSION AND FUTURE WORK

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